

## Fractional indices



1 Write the following in surd form:

a  $x^{\frac{1}{3}}$

b  $a^{\frac{3}{2}}$

c  $20^{\frac{5}{4}}$

d  $y^{-\frac{1}{2}}$

e  $3x^{\frac{1}{2}}$

f  $b^{2\frac{1}{3}}$

g  $y^{1.5}$

h  $5^{-\frac{4}{3}}$

2 Write the following as a single power of 2:

a  $\sqrt[3]{2}$

b  $\sqrt[5]{16}$

c  $\sqrt{32}$

d  $2\sqrt{2}$

e  $8\sqrt[3]{2}$

f  $2^a \times \sqrt{2}$

g  $\frac{8}{\sqrt{2}}$

h  $\frac{1}{\sqrt{8}}$

3 Write the following as a single power of 3:

a  $\sqrt{27}$

b  $3^x \times \sqrt[4]{3}$

c  $9\sqrt{3}$

d  $\frac{1}{\sqrt[3]{9^x}}$

4 Complete the following statement:

$$\begin{aligned}\sqrt{m^8} &= (m^8)^{\frac{1}{2}} \\ &= m^{8 \times \frac{1}{2}} \\ &= m^{\frac{1}{2}}\end{aligned}$$

5 Write the following in the form  $x^k$ , where  $k$  is rational:

a  $\sqrt{x}$

b  $\sqrt[6]{x}$

c  $\frac{1}{\sqrt{x}}$

d  $(\sqrt[3]{x})^2$

e  $\sqrt[4]{x^3}$

f  $(\sqrt[7]{x})^6$

g  $\sqrt[4]{x^8}$

h  $\frac{1}{\sqrt{x}}$

i  $\sqrt{\sqrt{x}}$

j  $\sqrt[3]{x^{6a}}$

k  $\frac{1}{\sqrt{x^{-3}}}$

l  $x\sqrt{x}$

m  $\frac{1}{\sqrt[4]{x^5}}$

n  $\sqrt[5]{\sqrt{x}}$

o  $x^2 \times \sqrt[3]{x}$

6 Consider the expression  $64^{\frac{2}{3}}$ .

a Complete the statement:  $64^{\frac{2}{3}} = (\sqrt[3]{64})^{\frac{1}{2}}$

b Hence, evaluate  $64^{\frac{2}{3}}$ .

7 Without using a calculator, evaluate the following:



a  $1^{\frac{1}{10}}$

b  $4^{\frac{3}{2}}$

c  $121^{\frac{1}{2}}$

d  $\sqrt[3]{27}$

e  $125^{\frac{1}{3}}$

f  $(-64)^{\frac{1}{3}}$

g  $64^{-\frac{1}{6}}$

h  $81^{-\frac{3}{4}}$

i  $\sqrt[3]{-64}$

j  $\left(\frac{9}{100}\right)^{\frac{1}{2}}$

k  $\left(64^{\frac{1}{9}}\right)^{\frac{9}{2}}$

l  $\left(\frac{8}{125}\right)^{\frac{2}{3}}$

m  $(-32)^{\frac{4}{5}}$

n  $\frac{64^{\frac{1}{3}}}{64^{\frac{2}{3}}}$

o  $\frac{1}{-\sqrt{169}}$

p  $\sqrt[3]{\frac{-64}{125}}$

q  $\frac{\sqrt[3]{40}}{\sqrt[3]{5}}$

r  $36^{\frac{1}{2}} - 32^{\frac{3}{5}}$

s  $1000^{\frac{8}{9}} \times 1000^{-\frac{5}{9}}$

t  $\sqrt[3]{\sqrt[4]{16} + \sqrt{625}}$

8 Use a calculator to evaluate the following, to two decimal places:

a  $10^{\frac{3}{2}}$

b  $4^{\frac{5}{3}}$

c  $\sqrt[4]{5^3}$

d  $(\sqrt[3]{12})^5$

9 Simplify the following expressions, giving your answers in index form. Assume that all variables represent positive numbers.

a  $(\sqrt{b})^8$

b  $\sqrt{m^6}$

c  $\sqrt[4]{a^5}$

d  $\frac{1}{\sqrt[5]{a^6}}$

e  $\sqrt[3]{x^6}$

f  $\sqrt[3]{m^3}$

g  $\sqrt{j^2 x^4}$

h  $\left(\sqrt[4]{x^5 y^3}\right)^{24}$

i  $(4a^8)^{\frac{1}{2}}$

j  $(625u^{16}v^{12})^{\frac{1}{2}}$

k  $8b^{\frac{3}{4}} \div 2b^{\frac{2}{3}}$

l  $y^3 \times \sqrt[3]{y}$

m  $\sqrt{(2x+9)^2}$

n  $\sqrt[3]{9^3 x^{18} y^{12}}$

o  $\left(\frac{x^{15}}{1024}\right)^{\frac{2}{5}}$

p  $\sqrt{\frac{36x^{18}}{y^{20}}}$

q  $\sqrt{16y^2 + 24y + 9}$

r  $\sqrt{x^2 y^2 + 18xy^2 + 81y^2}$

10 Simplify the following expressions, giving your answers in surd form. Assume that all variables represent positive numbers.

a  $\frac{\sqrt{x^2 + 5x + 6}}{\sqrt{x + 2}}$

b  $\frac{\sqrt[3]{x^2 + 9x + 20}}{\sqrt[3]{x + 4}}$

11 Describe how we could interpret the expression  $m^{\frac{q}{r}}$  in terms of powers and roots of  $m$ .

12 Determine whether the following statements accurately describe the meaning of the expression  $x^{-\frac{y}{z}}$ :



- a  $x^{-\frac{y}{z}}$  means we are raising  $x$  to the power of  $\frac{z}{y}$ , then taking the reciprocal of the result.
- b  $x^{-\frac{y}{z}}$  means we are taking the reciprocal of  $x$ , then raising the result to the power of  $\frac{y}{z}$ .
- c  $x^{-\frac{y}{z}}$  means we are taking the reciprocal of  $x$ , then raising the result to the power of  $\frac{z}{y}$ .
- d  $x^{-\frac{y}{z}}$  means we are raising  $x$  to the power of  $\frac{y}{z}$ , then taking the reciprocal of the result.

13 Is there a real number that equals  $\sqrt[4]{-16}$ ? Explain your answer.

14 Consider the expression  $m^5 \times m\sqrt{m}$ .

- a Express it in simplest index form.
- b Express it in surd form.

15 Consider the expression  $m^{\frac{1}{4}} = 8^{\frac{1}{2}}$ .

- a Complete the following working:

$$\begin{aligned} m^{\frac{1}{4}} &= (m^1)^{\frac{1}{2}} \\ &= \sqrt{1^{\frac{1}{2}}} \end{aligned}$$

- b Hence, or otherwise, solve the equation  $m^{\frac{1}{4}} = 8^{\frac{1}{2}}$ .

16 Consider the equation  $x^{\frac{2}{3}} = 9$ .

- a Complete the statement:  $(x^{\frac{2}{3}})^1 = x$
- b Hence, solve the equation  $x^{\frac{2}{3}} = 9$ .

17 Find the missing values in the following equations:

- a  $16^{\frac{3}{4}} = 64$
- b  $(1)^{\frac{2}{3}} = 25$
- c  $625^{\frac{3}{4}} = 125$
- d  $(1)^{\frac{4}{3}} = 81$

18 Solve the following equations for  $x$ :

- a  $x^{\frac{3}{4}} = 125$
- b  $(x - 4)^{0.5} = 4$
- c  $(2x - 1)^{\frac{2}{3}} = 9$
- d  $(4x + 3)^{-1.5} = 8$

19 Solve the following equation for  $k$ :

$$\sqrt[k]{y} \times \sqrt[k]{y} \times \sqrt[k]{y} = y^{\frac{1}{2}}$$

- 20** To evaluate  $81^{\frac{3}{2}}$  would it be more efficient to use the property  $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$ , or the property  $a^{\frac{m}{n}} = \sqrt[n]{a^m}$ . Explain your answer.



## Applications

- 21** The surface area  $A$  of a cube with a volume  $V$  is given by  $A = 6V^{\frac{2}{3}}$ .

- a** Find the surface area of a cube with a volume of  $8 \text{ cm}^3$ .
- b** Find the volume of a cube with a surface area of  $216 \text{ cm}^2$ .

- 22** The volume  $V$  of a sphere with surface area  $A$  is given by:

$$V = \frac{A^{\frac{3}{2}}}{6\sqrt{\pi}}$$

- a** Find the volume of a sphere with a surface area of  $50 \text{ cm}^2$ . Round your answer to the nearest cubic centimetre.
- b** Find the surface area of a sphere with a volume of  $125 \text{ cm}^3$ . Round your answer to the nearest square centimetre.

- 23** The volume  $V$  of a regular tetrahedron with edge length  $a$  is given by:

$$V = \frac{a^3}{6\sqrt{2}}$$

- a** Find the volume of a tetrahedron with a side length of  $2 \text{ cm}$ .
- b** Rearrange the formula to make  $a$  the subject.
- c** Hence, find the side length of a tetrahedon with a volume of  $72 \text{ cm}^3$ .

- 24** A company's net profit (in dollars) can be modelled by the equation:

$$P = 50n^{\frac{3}{4}} - 2500$$

Where  $n$  is the number of units sold.

- a** Find its net profit be in dollars if it manages to sell  $256$  units.
- b** Find the number of units needed to be sold to make a net profit of  $\$8300$ .

25 A company's net profit (in dollars) can be modelled by the equation:

$$P = 180n^{\frac{3}{5}} - 4000$$

Where  $n$  is the number of units sold.

- a Find its net profit in dollars if it manages to sell 32 units.
- b Find the number of units needed to be sold to make a net profit of \$45 000. Round to the nearest whole unit.